Dr. Chilperic Armel Foko Kuate

Computational Scientist | Multiscale Modeller | Scientific Software

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CORE STRENGTHS

- Multiscale simulation and numerical modelling
- Mechanistic, stochastic, and differential-equation systems
- Parameter estimation, optimisation, and sensitivity analysis
- Scientific software and reproducible computational workflows
- Interactive scientific and educational tool building
- Interdisciplinary collaboration and technical communication

TECHNICAL STACK

Programming
Python, Julia, R, MATLAB, JavaScript, HTML, CSS

Scientific Computing
NumPy, SciPy, CVODE, SUNDIALS, lmfit

Optimisation
CasADi, IPOPT, CMA-ES, JuMP, Pyomo, MEALPY, pyMetaheuristic

Analysis
SALib, Sobol, Morris, scikit-learn

Visualisation
matplotlib, Plotly, Bokeh, ggplot2, pandas

Workflow
Git, Jupyter, Linux, HPC, reproducible pipelines

EDUCATION

PhD, Quantitative and Theoretical Biology
Heinrich Heine University Düsseldorf 2019–2023

Master 2, Mathematics and Applications
University of Yaoundé I 2017–2019
Advanced numerical analysis, dynamical systems, inverse problems

MSc, Mathematical Sciences
AIMS Ghana 2015–2016
Applied probability, dynamical systems, scientific computing

BSc, Mathematics
University of Douala 2009–2012
Algebra, analysis, geometry

DEVELOPED WEB TOOLS

Foko Lab
Browser-based environment for ODE, optimisation, steady-state, and stochastic modelling workflows.

Open web tool

PROFESSIONAL SUMMARY

I am a computational scientist with strong training in pure and applied mathematics, mechanistic modelling, and scientific programming. I build multiscale simulation frameworks that connect governing principles to observable system behaviour, with particular strengths in differential-equation modelling, optimisation, uncertainty analysis, and reproducible scientific software. Across biology-focused research problems, I have repeatedly worked on questions whose structure is fundamentally physical, including transport, heat balance, nonlinear response, and constrained dynamics.

EXPERIENCE

Independent Research, Scientific Software, and Further Training Nov 2025 – Present
Düsseldorf, Germany

- Advancing two manuscripts on mechanistic modelling of plant adaptation across the C3–C4 continuum.
- Developing browser-based tools for scientific simulation and learning, including Foko Lab and Deutsch-WiPA 2026.
- Strengthening my profile in AI engineering, interactive technical tools, and professional German for applied research and software work.

Postdoctoral Researcher Nov 2022 – Oct 2025
CEPLAS / Institute of Computational Cell Biology, HHU Düsseldorf

- Built a multiscale thermo-hydraulic-biochemical simulation framework linking biochemical kinetics, heat transfer, hydraulics, and environmental forcing.
- Combined Sobol sensitivity analysis, large simulation ensembles, and CMA-ES optimisation to explore adaptive trait combinations under contrasting climates.
- Developed reproducible Python and Julia workflows for simulation, optimisation, visualisation, and systematic model exploration.
- Worked closely with experimental collaborators and supervised 2 MSc and 2 BSc research projects.

Marie Curie PhD Fellow Aug 2019 – Aug 2023
Institute of Quantitative and Theoretical Biology, HHU Düsseldorf

- Built mechanistic models of liver metabolism linking fatty-acid synthesis, oxidation, and degradation in interpretable nonlinear dynamical systems.
- Developed simulation, parameter-estimation, and model-analysis pipelines in Python and Julia for steady-state and time-course problems.
- Applied optimisation, identifiability analysis, sensitivity analysis, and uncertainty quantification to improve model robustness and interpretability.
- Contributed to enzyme-kinetic formalism for differentiable constraint-based metabolic models in cross-institutional work.

Research Fellow Nov 2017 – Mar 2019
AIMS Cameroon and AIMS Rwanda

- Developed stochastic and deterministic models of T-cell proliferation dynamics to extract interpretable parameters from CFSE-type experiments.
- Formulated a stochastic master-equation framework, derived analytical expressions for cell-count dynamics, and built a maximum-likelihood estimation workflow.
- Delivered more than 200 hours of teaching in mathematical modelling, algebra, stochastic processes, and Python programming.

SELECTED PUBLICATIONS

- Foko Kuate CA, Ortega F, Ebenhöh O. *A novel kinetic model of de novo fatty acid synthesis and its application to mitochondrial fatty acid oxidation disorders.* Bioscience Reports, 2022.
- Besançon M., Dourado H., Kratochvíl M., Foko Kuate CA, Trefois C., Gu W., Ebenhöh O. *Interrogating the Effect of Enzyme Kinetics on Metabolism Using Differentiable Constraint-Based Models.* Metabolic Engineering, 2022.
- Dourado H., Foko Kuate CA, Liebermeister W., Lercher M.J. *Growth Mechanics and the Emergence of Metabolic Oscillations in Growing Cells.* bioRxiv, 2025.

Deutsch-WiPA 2026

Interactive B1 to B2 German learning tool with conjugation practice, feedback loops, and error tracking.

Open web tool

LANGUAGES

French: Native
English: Fluent
German: B1+

INTERESTS

Scientific tool building, agriculture and farming, CrossFit, cooking, music, and long-term language learning.

REFERENCES

- Prof. Dr. Oliver Ebenhöh, Heinrich Heine University Düsseldorf
- Prof. Dr. Martin Lercher, Heinrich Heine University Düsseldorf
- Dr. Wilfred Ndifon, African Institute for Mathematical Sciences